

Loan Approval Prediction

Notebook link - [Loan Approval Prediction](#)

GitHub Repository - [Loan Approval github](#)

Project Overview -

This project builds a **Machine Learning model** to predict whether a loan application will be approved based on applicant information such as income, education, credit history, loan amount, and property area.

The project focuses on:

- Data preprocessing
 - Handling class imbalance
 - Model training & evaluation
 - Business-oriented insights for financial institutions
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Dataset -

- **Source:** Kaggle
- **Link:** [Loan-approval-prediction-dataset](#)

The dataset contains:

- Applicant demographic details
 - Income information
 - Loan amount & loan term
 - Credit history
 - Property area
 - Loan approval status
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Objectives -

- Perform data cleaning and preprocessing
- Handle missing values
- Encode categorical variables
- Scale numerical features

- Handle class imbalance using SMOTE
 - Compare multiple machine learning models
 - Evaluate models using classification metrics
 - Generate business recommendations
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Technologies Used -

- Python
 - Pandas & NumPy
 - Matplotlib & Seaborn
 - Scikit-learn
 - Imbalanced-learn (SMOTE)
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Data Cleaning & Preprocessing -

The following preprocessing steps were performed:

- Removed unnecessary columns (Loan_ID) •

Handled missing values using:

- i) Median imputation for numerical features ii) Most frequent imputation for categorical features

- Encoded categorical variables using One-Hot Encoding
 - Scaled numerical features using StandardScaler
 - Split data into training and testing sets
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Handling Class Imbalance -

The dataset contained imbalance in loan approval classes.

To solve this issue:

- **SMOTE (Synthetic Minority Oversampling Technique)** was applied
 - Minority class samples were synthetically generated
 - This improved model learning and reduced prediction bias
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SMOTE Balancing Output -

Before SMOTE -

Loan Approved (1): 337 Loan

Rejected (0): 154

After SMOTE -

Loan Approved (1): 337

Loan Rejected (0): 337

Exploratory Data Analysis (EDA) -

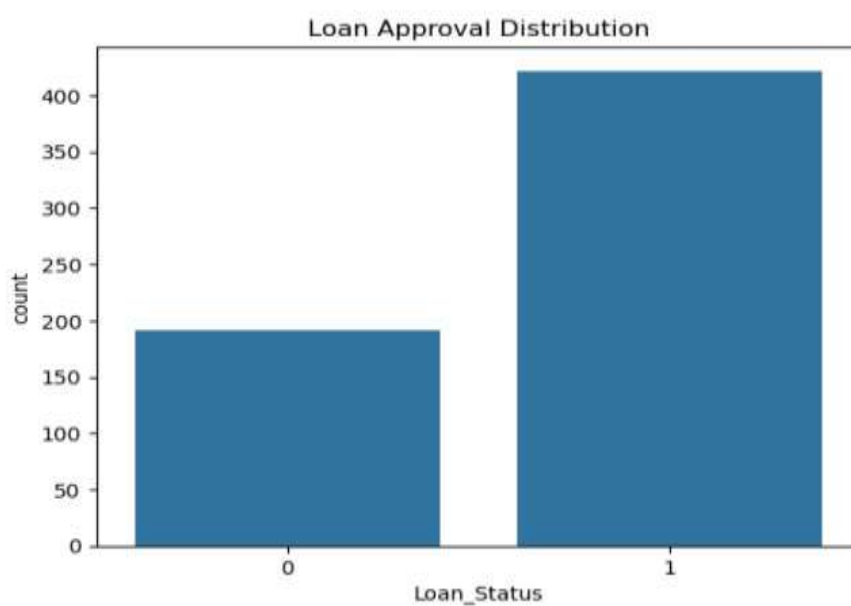
Key Analysis Performed:

- Loan approval distribution
- Credit history analysis
- Income distribution
- Loan amount trends
- Property area impact
- Feature relationships

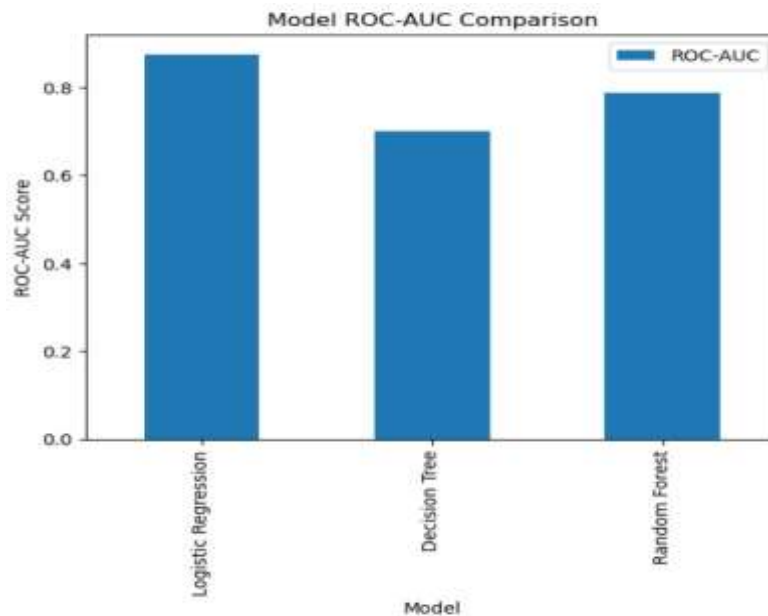
Visualizations -

The following charts were created:

- Loan approval distribution chart –



- ROC-AUC comparison chart



- Feature importance output

Top Important Features:

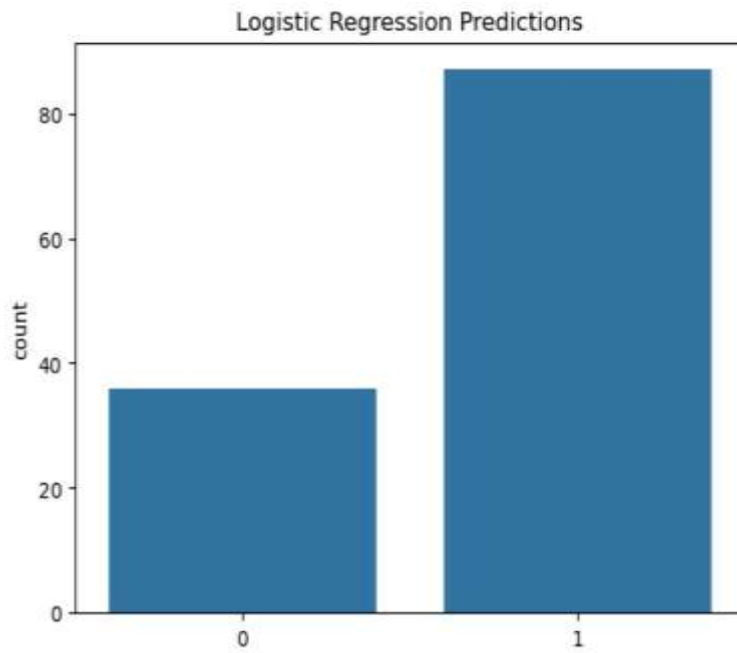
	Feature	Coefficient
4	num_Credit_History	1.264772
18	cat_Property_Area_Semiurban	0.485655
11	cat_Dependents_2	0.276412
13	cat_Education_Graduate	0.176153
0	num_ApplicantIncome	0.144487
8	cat_Married_Yes	0.143198
12	cat_Dependents_3+	0.132878
5	cat_Gender_Female	0.074236
15	cat_Self_Employed_No	0.025427
3	num_Loan_Amount_Term	-0.015256

Machine Learning Models -

The following supervised learning models were trained:

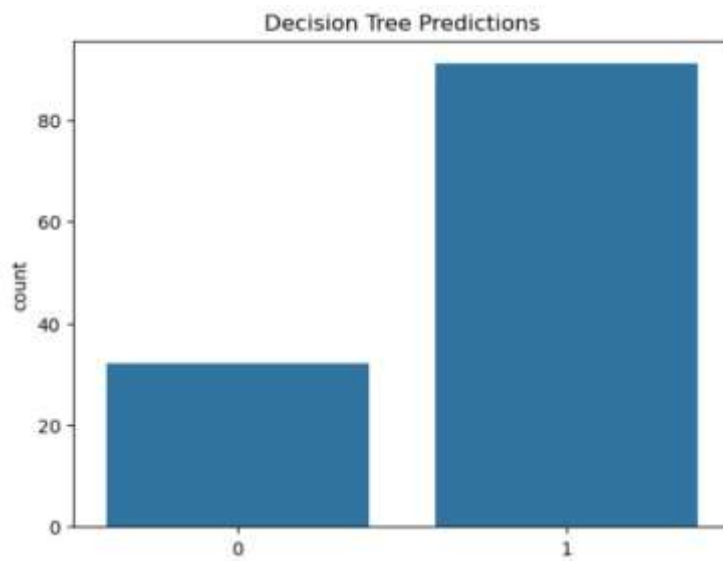
1. Logistic Regression -

- Simple and interpretable model
- Performs well on structured financial data



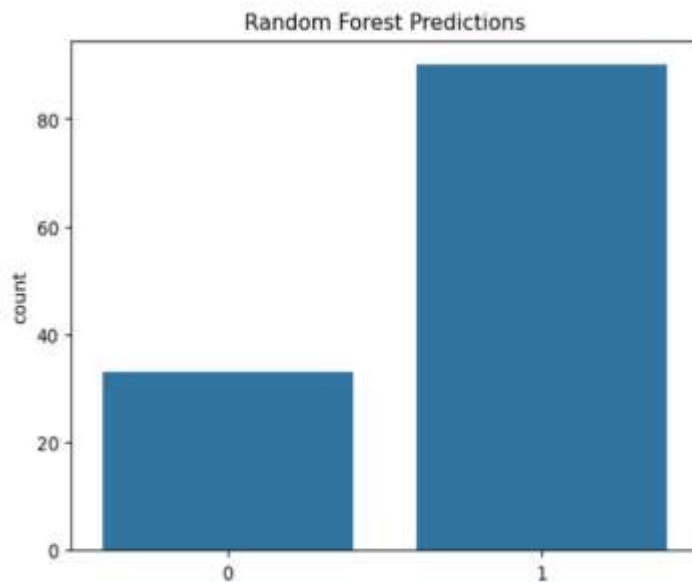
2. Decision Tree -

- Captures non-linear decision boundaries
- Easy to visualize



3. Random Forest -

- Ensemble tree-based model
- Reduces overfitting and improves prediction stability



Model Evaluation Metrics -

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

Model Comparison Table -

Model	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.862	0.882	0.872	0.876
Decision Tree	0.802	0.859	0.830	0.700
Random Forest	0.833	0.882	0.857	0.788

Best Model -

Logistic Regression -

Logistic Regression achieved the best overall performance based on:

- Highest ROC-AUC score
- Strong F1-score
- Better interpretability

- Stable predictions

This makes it suitable for financial institutions where explainability is important.

Feature Importance -

Important factors influencing loan approval:

- Credit History
- Applicant Income
- Property Area • Education
- Marital Status

Applicants with strong credit history and stable income had higher approval probability.

Business Interpretation -

Key Insights -

- Applicants with good credit history are more likely to receive loan approval.
- Higher income positively impacts approval probability.
- Balanced loan amount and repayment capacity reduce lending risk.
- Property area and education level also influence approvals. **Business Benefits -**

Banks and financial institutions can use this model to:

- Speed up loan approval decisions
 - Reduce manual verification effort
 - Minimize risky loan approvals
 - Improve customer screening
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Suggested Deployment Threshold -

Instead of using the default probability threshold of **0.50**, financial institutions can use:

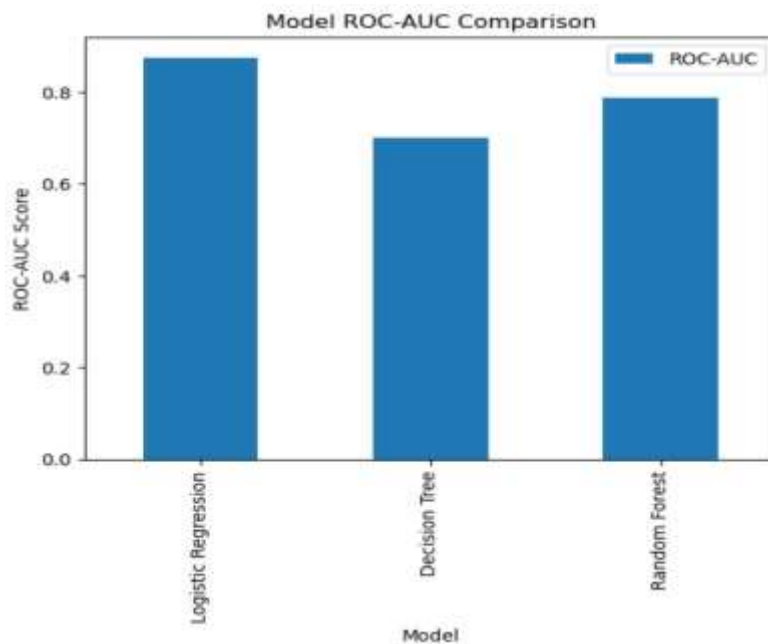
Recommended Threshold: 0.60

Benefits:

- Reduces risky approvals
- Minimizes default probability

- Improves lending reliability

ROC-AUC Comparison Chart –



Conclusion -

This project successfully developed a complete supervised machine learning pipeline for loan approval prediction.

The workflow included:

- Data preprocessing
- Missing value handling
- Feature encoding • Feature scaling
- SMOTE balancing
- Model training & comparison
- Business interpretation

Among all models, **Logistic Regression** provided the best balance between performance and interpretability, making it the preferred model for deployment in real-world financial systems.

Submitted By - Sargam

Ghagre Domain – Data

Science